Reg. No. | | | | | | | | | | | [14AE61]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE–2019

AEROSPACE DESIGN-I

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full questions choosing at least two from each part.
2. Draw sketches wherever required.
3. Any missing data maybe suitably assumed and stated.
4. Derive the formulae wherever necessary and substitute the given parameters for calculation.

PART - A

1. a. What is Reynold's Number? Explain its significance in aircraft design. **04**

b The flight test data for a given airplane refer to a level-flight maximum-velocity run made at an altitude which simultaneously corresponded to a pressure altitude of 9150 m and a density altitude of 8700 m. Using standard atmospheric data (given below) calculate the temperature of the air at the altitude at which the airplane was flying for the test. **07**

Standard values at altitude								
Alt. (m)	T (K)	θ	a (m/s)	p (Pa)	δ	ρ (kg/m^3)	σ	μ/μ_{sl}
8,000	236.15	0.8195	308.06	35,599	0.3513	0.5252	0.4287	0.8532
8,500	232.90	0.8083	305.93	33,099	0.3267	0.4951	0.4042	0.8436
9,000	229.65	0.7970	303.79	30,742	0.3040	0.4663	0.3807	0.8339
9,500	226.40	0.7857	301.63	28,523	0.2815	0.4389	0.3583	0.8241

c. What is drag? List different types of drag that can be experienced by an aircraft. **03**

2. a. With a neat sketch describe the layout of a turbofan engine. Write the expression for the thrust produced by the turbofan engine. **06**

b. What are the key parameters that must be considered for engine selection? **05**

c. Define Propulsive efficiency, Thermal efficiency and Overall efficiency. **03**

3. a. Using first principles (longitudinal and rotational force equations) derive the equation for steady level cruise flight. **07**

b. Calculate the thrust required at sea level for an aircraft assuming the following: Wing span= 35.8 ft, Wing area= 174 ft2, Normal gross weight = 2950 lb, Specific fuel consumption= 0.45 lb/hp-h, Parasite drag coefficient= 0.025, Oswald efficiency factor= 0.8, Propeller efficiency= 0.8 and V_∞= 200 ft/sec. **07**

4. a. List seven important performance and mission requirements of an airplane, at the stage of conceptual design. Explain how each of the requirements affects the design parameters for the aircraft. **07**

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b. Draw and explain mission profile for a long range stealth bomber aircraft with a neat sketch. **07**

PART - B

5. a. Derive the expression for maximum power-off glide range for a fixed wing aircraft. **04**

b. Using thrust vs. velocity charts explain the terms Absolute Ceiling and Service Ceiling. **05**

Consider a light, single-engine airplane. If the maximum gross weight of the airplane is 7780 N, the wing area is 16.6 m^2, and the maximum lift coefficient is 2.1 with flaps down, calculate the stalling speed at sea level. **05**

6. a. What is the importance of Thrust to Weight ratio in aircraft design? Explain how thrust matching is essential for aircraft design. **07**

b. What is the taper ratio of an aircraft wing? How does the taper ratio affect lift distribution? What is the effect of taper ratio on wing stall? **07**

7. a. Explain in detail about VTOL propulsion considerations. **08**

b. What is static stability? Establish the condition for longitudinal static stability of an aircraft (neglecting the tail). **06**

8. a. Explain the V-N diagram of an aircraft and detail all its salient points. **07**

b. Write shot notes on:

i. Control Surface Sizing during conceptual aircraft design **07**

ii. Canard tail configuration